YuEEG REV D / ISOLATED USB + WIFI / N16R8 / SYMMETRIC AFE EXPANSION

01 / Electrode inputs CH1–CH4

File: inputs_1_4.kicad_sch

01 / Electrode inputs CH5–CH8

File: inputs_5_8.kicad_sch

02 / ADS1299 digital, bias, reference and decoupling

File: afe_core.kicad_sch

03 / ESP32-S3 wireless acquisition controller

File: mcu_radio.kicad_sch

04 / Battery input and independent analog–digital supplies

File: power.kicad_sch

05 / Bipolar analog supply

File: power_bipolar.kicad_sch

06 / USB-C standalone 2S charger and battery NTC

File: battery_charger.kicad_sch

07 / Raw 2S cell protection and fail-safe acquisition switch

File: battery_protection.kicad_sch

08 / Cell balancing with raw-domain UV qualification

File: battery_balance.kicad_sch

09 / Isolated USB data and conditional isolated power

File: isolated_usb.kicad_sch

10 / USB SYS power path with delayed battery interlock

File: usb_system_power.kicad_sch

11 / Symmetric AFE insertion and status interfaces

File: afe_expansion.kicad_sch

Engineering prototype: isolation components and layout require system safety, leakage, dielectric and EMC validation.
USB data and WiFi acquisition run together. POWER OFF for charging; see REV_D_REVIEW.md.

ISOLATED USB PROTOTYPE – SYSTEM VALIDATION REQUIRED

YuEEG

Sheet: /
File: YuEEG_8CH_RevD.kicad_sch

Title: YuEEG 8CH – wireless EEG acquisition

Size: A3	Date: 2026-10-25	**Rev: D**
KiCad E.D.A. 9.0.5		Id: 1/13

01 / ELECTRODE INPUTS CH1–CH4

Referential EEG: one SIGNAL terminal per channel. Shared REF –> all N input networks. BIAS is a separate driven electrode.

CH1 | 2 x 4.99k per lead; 1nF differential; 47pF matched COG common–mode

J101
CH1 EEG
E1P
1

R111
4.99k 0.1%
E1P
CLAMP1P
BASE_RET_1P
AIN1P

R112
4.99k 0.1%
AIN1P

R113
4.99k 0.1%
E1N
CLAMP1N
BASE_RET_1N
AIN1N

R114
4.99k 0.1%
AIN1N

R115
OR REF SELECT
REF_RAW
E1N

CH2 | 2 x 4.99k per lead; 1nF differential; 47pF matched COG common–mode

J102
CH2 EEG
E2P
1

R121
4.99k 0.1%
E2P
CLAMP2P
BASE_RET_2P
AIN2P

R122
4.99k 0.1%
AIN2P

R123
4.99k 0.1%
E2N
CLAMP2N
BASE_RET_2N
AIN2N

R124
4.99k 0.1%
AIN2N

R125
OR REF SELECT
REF_RAW
E2N

CH3 | 2 x 4.99k per lead; 1nF differential; 47pF matched COG common–mode

J103
CH3 EEG
E3P
1

R131
4.99k 0.1%
E3P
CLAMP3P
BASE_RET_3P
AIN3P

R132
4.99k 0.1%
AIN3P

R133
4.99k 0.1%
E3N
CLAMP3N
BASE_RET_3N
AIN3N

R134
4.99k 0.1%
AIN3N

R135
OR REF SELECT
REF_RAW
E3N

CH4 | 2 x 4.99k per lead; 1nF differential; 47pF matched COG common–mode

J104
CH4 EEG
E4P
1

R141
4.99k 0.1%
E4P
CLAMP4P
BASE_RET_4P
AIN4P

R142
4.99k 0.1%
AIN4P

R143
4.99k 0.1%
E4N
CLAMP4N
BASE_RET_4N
AIN4N

R144
4.99k 0.1%
AIN4N

R145
OR REF SELECT
REF_RAW
E4N

D111
BAV199
–2V5A
A1
K1/A2
K2
+2V5A
3
CLAMP1P

D112
BAV199
–2V5A
A1
K1/A2
K2
+2V5A
3
CLAMP1N

D121
BAV199
–2V5A
A1
K1/A2
K2
+2V5A
3
CLAMP2P

D122
BAV199
–2V5A
A1
K1/A2
K2
+2V5A
3
CLAMP2N

D131
BAV199
–2V5A
A1
K1/A2
K2
+2V5A
3
CLAMP3P

D132
BAV199
–2V5A
A1
K1/A2
K2
+2V5A
3
CLAMP3N

D141
BAV199
–2V5A
A1
K1/A2
K2
+2V5A
3
CLAMP4P

D142
BAV199
–2V5A
A1
K1/A2
K2
+2V5A
3
CLAMP4N

C111
47p C0G 1%
AIN1P
GND

C112
47p C0G 1%
AIN1N
GND

C113
1n C0G 2%
AIN1P
AIN1N

C121
47p C0G 1%
AIN2P
GND

C122
47p C0G 1%
AIN2N
GND

C123
1n C0G 2%
AIN2P
AIN2N

C131
47p C0G 1%
AIN3P
GND

C132
47p C0G 1%
AIN3N
GND

C133
1n C0G 2%
AIN3P
AIN3N

C141
47p C0G 1%
AIN4P
GND

C142
47p C0G 1%
AIN4N
GND

C143
1n C0G 2%
AIN4P
AIN4N

U1A
ADS1299PAG
AIN1P
16
IN1P
AIN1N
15
IN1N
AIN2P
14
IN2P
AIN2N
13
IN2N
AIN3P
12
IN3P
AIN3N
11
IN3N
AIN4P
10
IN4P
AIN4N
9
IN4N

Front–end RC is RF/alias prefiltering, not the 0.5–80Hz digital bandpass.
BAV199 steers to +/–2.5V; rail steering, NOT certified patient protection or a 0.3V hard clamp.
Reference leads are also current–limited through the N–side resistors.
No electrode is connected directly to board GND.

ISOLATED USB PROTOTYPE – SYSTEM VALIDATION REQUIRED
YuEEG

Sheet: /01 / Electrode inputs CH1–CH4/		
File: inputs_1_4.kicad_sch		
Title: 01 / Electrode inputs CH1–CH4		
Size: A3	Date: 2026–10–26	**Rev: D**
KiCad E.D.A. 9.0.5		Id: 2/13

01 / ELECTRODE INPUTS CH5–CH8

Referential EEG: one SIGNAL terminal per channel. Shared REF –> all N input networks. BIAS is a separate driven electrode.

CH5 | 2 x 4.99k per lead; 1nF differential; 47pF matched COG common–mode

J105
CH5 EEG
E5P
1

R151
4.99k 0.1%
E5P
CLAMP5PBASE_RET_5P
AIN5P

R152
4.99k 0.1%
AIN5P

R153
4.99k 0.1%
E5N
CLAMP5NBASE_RET_5N
AIN5N

R154
4.99k 0.1%
AIN5N

R155
OR REF SELECT
REF_RAW
E5N

D151
BAV199
–2V5A
A1
K2
+2V5A
K1/A2
3
CLAMP5P

D152
BAV199
–2V5A
A1
K2
+2V5A
K1/A2
3
CLAMP5N

AIN5P
C151
47p C0G 1%
GND

AIN5N
C152
47p C0G 1%
GND

AIN5P
C153
1n C0G 2%
AIN5N

CH6 | 2 x 4.99k per lead; 1nF differential; 47pF matched COG common–mode

J106
CH6 EEG
E6P
1

R161
4.99k 0.1%
E6P
CLAMP6PBASE_RET_6P
AIN6P

R162
4.99k 0.1%
AIN6P

R163
4.99k 0.1%
E6N
CLAMP6NBASE_RET_6N
AIN6N

R164
4.99k 0.1%
AIN6N

R165
OR REF SELECT
REF_RAW
E6N

D161
BAV199
–2V5A
A1
K2
+2V5A
K1/A2
3
CLAMP6P

D162
BAV199
–2V5A
A1
K2
+2V5A
K1/A2
3
CLAMP6N

AIN6P
C161
47p C0G 1%
GND

AIN6N
C162
47p C0G 1%
GND

AIN6P
C163
1n C0G 2%
AIN6N

CH7 | 2 x 4.99k per lead; 1nF differential; 47pF matched COG common–mode

J107
CH7 EEG
E7P
1

R171
4.99k 0.1%
E7P
CLAMP7PBASE_RET_7P
AIN7P

R172
4.99k 0.1%
AIN7P

R173
4.99k 0.1%
E7N
CLAMP7NBASE_RET_7N
AIN7N

R174
4.99k 0.1%
AIN7N

R175
OR REF SELECT
REF_RAW
E7N

D171
BAV199
–2V5A
A1
K2
+2V5A
K1/A2
3
CLAMP7P

D172
BAV199
–2V5A
A1
K2
+2V5A
K1/A2
3
CLAMP7N

AIN7P
C171
47p C0G 1%
GND

AIN7N
C172
47p C0G 1%
GND

AIN7P
C173
1n C0G 2%
AIN7N

CH8 | 2 x 4.99k per lead; 1nF differential; 47pF matched COG common–mode

J108
CH8 EEG
E8P
1

R181
4.99k 0.1%
E8P
CLAMP8PBASE_RET_8P
AIN8P

R182
4.99k 0.1%
AIN8P

R183
4.99k 0.1%
E8N
CLAMP8NBASE_RET_8N
AIN8N

R184
4.99k 0.1%
AIN8N

R185
OR REF SELECT
REF_RAW
E8N

D181
BAV199
–2V5A
A1
K2
+2V5A
K1/A2
3
CLAMP8P

D182
BAV199
–2V5A
A1
K2
+2V5A
K1/A2
3
CLAMP8N

AIN8P
C181
47p C0G 1%
GND

AIN8N
C182
47p C0G 1%
GND

AIN8P
C183
1n C0G 2%
AIN8N

U1B
ADS1299PAG
AIN5P
8
IN5P
AIN5N
7
IN5N
AIN6P
6
IN6P
AIN6N
5
IN6N
AIN7P
4
IN7P
AIN7N
3
IN7N
AIN8P
2
IN8P
AIN8N
1
IN8N

Front–end RC is RF/alias prefiltering, not the 0.5–80Hz digital bandpass.
BAV199 steers to +/–2.5V; rail steering, NOT certified patient protection or a 0.3V hard clamp.
Reference leads are also current–limited through the N–side resistors.
No electrode is connected directly to board GND.

ISOLATED USB PROTOTYPE – SYSTEM VALIDATION REQUIRED
YuEEG

Sheet: /01 / Electrode inputs CH5–CH8/		
File: inputs_5_8.kicad_sch		
Title: 01 / Electrode inputs CH5–CH8		
Size: A3	Date: 2026–10–26	**Rev: D**
KiCad E.D.A. 9.0.5		Id: 3/13

02 / ADS1299 DIGITAL, BIAS, REFERENCE AND DECOUPLING

U1C
ADS1299PAG

ADS_MOSI 34 DIN
ADS_SCLK 40 SCLK
ADS_CS 39 CS
ADS_RESET 36 RESET
ADS_PWDN 35 PWDN
ADS_START 38 START
CLKSEL 52 CLKSEL
37 CLK
GND 41 DAISY_IN
GND 31 RESV1

DOUT 43 ADS_MISO
DRDY 47 ADS_DRDY
GPIO1 42 GPIO1_PD
GPIO2 44 GPIO2_PD
GPIO3 45 GPIO3_PD
GPIO4 46 GPIO4_PD

+3V3 R201 10k ADS_CS
GND R202 10k ADS_RESET
GND R203 10k ADS_PWDN
+3V3 R204 10k CLKSEL

ADS_START R205 10k GND
GPIO1_PD R206 10k GND
GPIO2_PD R207 10k GND
GPIO3_PD R208 10k GND
GPIO4_PD R209 10k GND

Internal 2.048MHz clock: CLKSEL pulled high; CLK output disabled.
Use SPI MODE1. DRDY -> ISR notification -> SPI/DMA task -> packet FIFO.
RESET/PWDN default LOW. Release PWDN after rails settle; hold RESET.
Wait 10s initially; verify VCAP1–AVSS >1.1V BEFORE releasing RESET.

U1E
ADS1299PAG

+2V5A 19 AVDD
+2V5A 21 AVDD
+2V5A 22 AVDD
+2V5A 56 AVDD
+2V5A 59 AVDD
+2V5A 54 AVDD1
+3V3 48 DVDD
+3V3 50 DVDD

AVSS 20 −2V5A
AVSS 23 −2V5A
AVSS 32 −2V5A
AVSS 57 −2V5A
AVSS 58 −2V5A
AVSS1 53 −2V5A
DGND 33 GND
DGND 49 GND
DGND 51 GND
VCAP1 28 VCAP1
VCAP2 30 VCAP2
VCAP3 55 VCAP3
VCAP4 26 VCAP4
NC 27
NC 29

C210..215: AVDD to AVSS at EACH analog supply pair.
C216/217: one at EACH digital supply pin.
Keep VCAP and VREF loops short.

U1D
ADS1299PAG

17 SRB1
18 SRB2
−2V5A 60 BIASREF
BIAS_INV 61 BIASINV
−2V5A 62 BIASIN

BIASOUT 63 BIAS_OUT
VREFP 24 VREFP
VREFN 25 −2V5A
RESERVED 64

R210 1M 1% BIAS_OUT BIAS_INV
R211 499k 1% BIAS_OUT BIAS_LIMIT
R212 499k 1% BIAS_LIMIT BIAS_ELECTRODE

C201 1.5n C0G BIAS_OUT BIAS_INV

J201
REF / BIAS
REF_RAW 1
BIAS_ELECTRODE 2

+2V5A C210 1u 10V X7R −2V5A
+2V5A C211 1u 10V X7R −2V5A
+2V5A C212 1u 10V X7R −2V5A

VREFP C202 10u 10V X7R −2V5A
VREFP C203 100n 10V −2V5A

+2V5A C213 1u 10V X7R −2V5A
+2V5A C214 1u 10V X7R −2V5A
+2V5A C215 1u 10V X7R −2V5A

+3V3 C216 1u 10V X7R GND
+3V3 C217 1u 10V X7R GND

BIASREF_INT=1 -> internal near–0V midpoint. SRB1/SRB2 unused.
Use CHnSET=0x60 and MISC1=0x00 (external N inputs).
Start with BIAS_SENSP=0xFF, BIAS_SENSN=0x00; validate loop stability.
BIAS feedback 1M || 1.5nF is a TI example, not a guaranteed stable loop.
Lead–off excitation disabled in baseline recording; enable only in a separate test mode.

D801
BAV199 BIAS CLAMP
−2V5A 1 A1
K2 2 +2V5A
K1/A2
3 BIAS_LIMIT

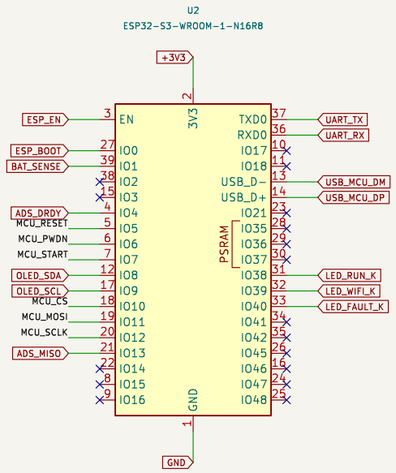
VCAP1 + C220 100u 10V −2V5A
VCAP2 C221 1u 10V −2V5A
VCAP3 C222 1u 10V −2V5A

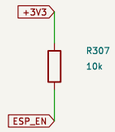
VCAP3 C223 100n 10V −2V5A
VCAP4 C224 1u 10V −2V5A

ISOLATED USB PROTOTYPE — SYSTEM VALIDATION REQUIRED
YuEEG

Sheet: /02 / ADS1299 digital, bias, reference and decoupling/		
File: afe_core.kicad_sch		
Title: 02 / ADS1299 digital, bias, reference and decoupling		
Size: A3	Date: 2026-10-26	**Rev: D**
KiCad E.D.A. 9.0.5		Id: 4/13

03 / ESP32-S3 WIRELESS ACQUISITION CONTROLLER











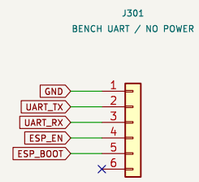




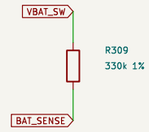


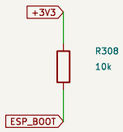






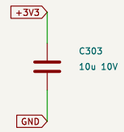
J301 is 3.3V UART, not RS-232.
No USB or charging power path is fitted.
Disconnect ALL electrodes before attaching
UART, debugger, scope or external instruments.

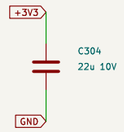


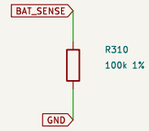


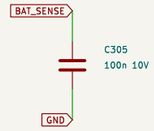












VBAT/4.3 -> GPIO1 ADC1. Max 8.4V -> 1.95V.
Calibrate ADC and stop acquisition below 6.4V.
Pack protection is external; this divider is NOT a BMS.

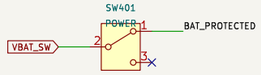
GPIO map: IO4 DRDY / IO5 RESET / IO6 PWDN / IO7 START / IO10 CS / IO11 MOSI / IO12 SCLK / IO13 MISO.
Boot strap pins IO3, IO45, IO46 are unused. Choose N8 (no PSRAM), not an arbitrary module variant.
Transmit int24 samples + sequence counter + sample-rate/gain metadata; target 500SPS, 24x.
Do not print volts from the interrupt. This hardware requires new pin mapping and firmware integration.
RF module antenna at PCB edge; preserve Espressif antenna keep-out on all layers.

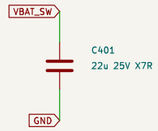
ISOLATED USB PROTOTYPE – SYSTEM VALIDATION REQUIRED
YuEEG
Sheet: /03 / ESP32-S3 wireless acquisition controller/
File: mcu_radio.kicad_sch
Title: 03 / ESP32-S3 wireless acquisition controller
Size: A3 | Date: 2026-10-26 | **Rev: D**
KiCad E.D.A. 9.0.5 | Id: 5/13

04 / BATTERY INPUT AND INDEPENDENT ANALOG–DIGITAL SUPPLIES

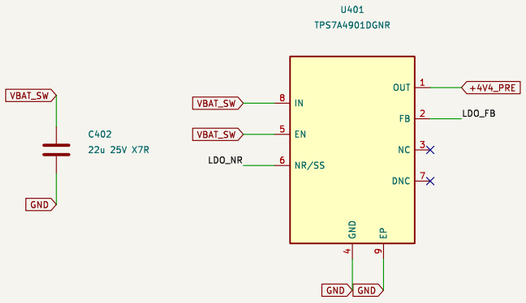


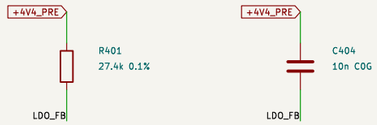


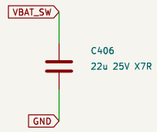


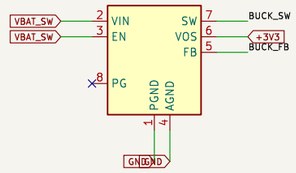


Onboard 2S charger/protection/balancer: see battery–management sheets.
J401 now has B+, MID, B–; connect a matched 2S pack plus a separate insulated NTC.
Isolated USB may supply acquisition via delayed SYS path. POWER OFF for charging. System isolation validation pending.

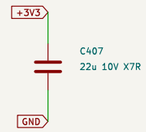


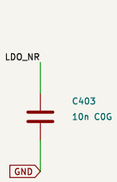


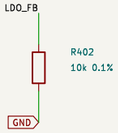


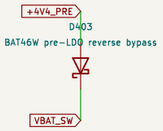


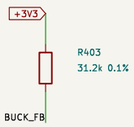


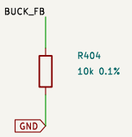












Pre–regulator (NOT ADS AVDD): 1.185 x (1 + 27.4k/10k) = 4.432V nominal.
22uF nominal input/output; >=10uF effective after DC bias.
Feeds TPS73225 and TPS60403 on bipolar supply sheet. LDO EP to GND. Keep digital switching return away from AFE.















Digital: 0.8 x (1 + 31.2k/10k) = 3.296V nominal.
VOS senses output AFTER inductor. 1A rated converter.
Minimize VIN–cap/SW/PGND hot loop; validate Wi–Fi transients.

ISOLATED USB PROTOTYPE – SYSTEM VALIDATION REQUIRED

YuEEG

Sheet: /04 / Battery input and independent analog–digital supplies/
File: power.kicad_sch

Title: 04 / Battery input and independent analog–digital supplies

Size: A3	Date: 2026–10–26	**Rev: D**
KiCad E.D.A. 9.0.5		Id: 6/13

BIPOLAR ANALOG SUPPLY: +2.5V / −2.5V; independent 3.3V digital ground reference

RESET/PWDN default LOW. After rails settle: PWDN HIGH, keep RESET LOW.
Wait >=tPOR AND VCAP1−AVSS >1.1V; use 10s initial bench delay.
100uF / internal 9.8k means seconds, NOT 200ms. Verify with scope.
Then RESET HIGH; wait >=18 CLK; SDATAC/configure/read back/start.
No automatic analog−rail/VCAP monitor is fitted; qualify timing on bench.

+4V4_PRE
C410
1u 10V X7R
GND

U403
TPS73225DBVR
+4V4_PRE
+4V4_PRE
IN
EN
OUT
NR
GND
+2V5A
POS_NR
GND

+2V5A
C412
10u 10V X7R
GND

+2V5A
R410
4.99k 1%
GND

POS_NR
C411
10n COG
GND

C414
2.2u 10V X7R
CFLY_PLUS
CFLY_MINUS

NEG_RAW
D402
BAT46W reverse−current bypass
−2V5A

+4V4_PRE
C413
2.2u 10V X7R
GND

U404
TPS60403DBVR
+4V4_PRE
IN
OUT
CFLY+
GND
CFLY−
NEG_RAW
CFLY_PLUS
CFLY_MINUS
GND

GND
C415
10u 10V X7R
NEG_RAW

U405
TPS72325DBVR
NEG_RAW
+3V3
IN
EN
OUT
NR
GND
−2V5A
NEG_NR
GND

GND
C417
10u 10V X7R
−2V5A

GND
R411
4.99k 1%
−2V5A

NEG_NR
C416
10n COG
GND

GND
C418
1u 10V X7R
NEG_RAW

TP405
+2V5A
+2V5A

TP406
−2V5A
−2V5A

TP407
NEG_RAW
NEG_RAW

4.43V pre−regulator −> positive LDO: 250kHz inverter −> negative low−noise LDO.
Keep CFLY/input/output capacitor loops at U404. Do not route switching nodes into the AFE.
TPS723 EN is +3V3 (not +4V4_PRE); 10nF NR to GND. D402 bypasses reverse current at power−down.
4.99k rail bleeders provide ~0.5mA load/discharge; they do not absorb arbitrary electrode faults.

ISOLATED USB PROTOTYPE − SYSTEM VALIDATION REQUIRED
YuEEG
Sheet: /05 / Bipolar analog supply/
File: power_bipolar.kicad_sch
Title: 05 / Bipolar analog supply and power−up requirements
Size: A3 | Date: 2026−10−26 | **Rev: D**
KiCad E.D.A. 9.0.5 | Id: 7/13

USB-C 5V INPUT / 2S Li-ion CHARGER / 300mA / 8.2V CONSERVATIVE CV

F602
1A hold PTC

C601
4.7u 16V X7R

C602
47u 16V X7R

C605
47u 16V X7R

C606
47u 16V X7R

C607
22u 25V X7R

J601
ISOLATED USB-C DATA + CHARGE

D601
SMF5.0A

C603
100n 16V

U601
BQ25886RGER

C608
47n 25V X7R

C604
22u 16V X7R

L601
1uH SRP5030T-1R0M / Isat11A

R601
10k 1%

R602
10k 1%

R603
1.50k 1%

R604
1.14k 1%

R605
9.31k 0.1%

R606
365k 0.1%

#FLG601
PWR_FLG

J602
CELL NTC / 103AT-2 10k

C609
100n 16V

R609
2.2k

LED601
CHARGING

D602
PESD5V0U1BAF

D603
PESD5V0U1BAF

D604
PESD5V0S1BA

D605
PESD5V0S1BA

R607
100k 1%

R608
100k 1%

#FLG602
PWR_FLG

USB HOST_GND and HOST_RAW are isolated from system GND. See isolated_usb sheet.
Charger DP/DM are shorted ONLY locally; USB data uses ADuM4160 to ESP32-S3.
Charging input is enabled only for Type-C 1.5A/3A advertisement; legacy ports support battery-powered data.
8.2V / nominal 299mA; independent insulated NTC. POWER ON holds CE high: acquisition and charge are separate modes.
USB power can supply the board through charger SYS and a delayed interlocked path; no unprotected BAT bypass.

ISOLATED USB PROTOTYPE – SYSTEM VALIDATION REQUIRED

YuEEG

Sheet: /06 / USB-C standalone 2S charger and battery NTC/
File: battery_charger.kicad_sch

Title: 06 / USB-C standalone 2S charger and battery NTC

Size: A3	Date: 2026-10-26	**Rev: D**
KiCad E.D.A. 9.0.5		Id: 8/13

RAW CELL HARNESS: B+ / MID / B− ; NEVER SHORT RAW B− TO SYSTEM GND

J401
2S RAW: B+ / MID / B−
RAW_BPLUS
RAW_MID
RAW_BMINUS

F601
1A fast fuse / 32V
RAW_BPLUS
PACK_POS

U602
S-8252AAY-M6T1U
PROT_VDD
PROT_VC
PROT_VM
VDD
VC
VM
DO
CO
VSS
DSG_GATE
CHARGE_GATE
RAW_BMINUS

U604
TPS3700DDCR RUN WINDOW
PACK_POS
RUN_UV
RUN_OV
VDD
INA+
INB−
OUTA
OUTB
GND
RUN_BASE
RUN_BASE
GND

C640
100n 16V
PACK_POS
GND

R620
470R 1%
RAW_BPLUS
PROT_VDD

R621
470R 1%
RAW_MID
PROT_VC

C620
100n 16V
PROT_VDD
RAW_BMINUS

C621
100n 16V
PROT_VC
RAW_BMINUS

R622
2.00k 1%
GND
PROT_VM

R640
1.58M 0.1%
PACK_POS
RUN_UV

R641
100k 0.1%
RUN_UV
GND

R642
2.05M 0.1%
PACK_POS
RUN_OV

R643
100k 0.1%
RUN_OV
GND

Q601
CSD17308Q3 DSG
DSG_SOURCE
DSG_SOURCE
DSG_SOURCE
S1
S2
S3
D5-8_EP
G
BMS_DRAINS
DSG_GATE

Q602
CSD17308Q3 CHG
GND
GND
GND
S1
S2
S3
D5-8_EP
G
BMS_DRAINS
CHARGE_GATE

R623
0.010R 1% 0.5W
RAW_BMINUS
DSG_SOURCE

C641
10n C0G
RUN_UV
GND

R644
62k 1%
PACK_POS
RUN_BASE

R645
10k
RUN_BASE
RUN_NPN_BASE

R646
1M
RUN_NPN_BASE
GND

Q605
MMBT3904 ENABLE
RUN_PULLDOWN
RUN_NPN_BASE
GND

Q606
MMBT3904 USB INHIBIT
RUN_BASE
USB_INHIBIT_B
GND

R647
47k
USB_RAW
USB_INHIBIT_B

R648
1M
USB_INHIBIT_B
GND

D606
BZT52C8V2S
DSG_GATE
DSG_SOURCE

D607
BZT52C8V2S
CHARGE_GATE
GND

R650
100k
RUN_GATE
RUN_PULLDOWN

Q603
AO3401A BACK-TO-BACK
PACK_POS
RUN_GATE
RUN_SOURCES

Q604
AO3401A BACK-TO-BACK
ACQ_INPUT
RUN_GATE
RUN_SOURCES

R649
1M
RUN_SOURCES
RUN_GATE

C642
10n 25V
RUN_GATE
RUN_SOURCES

R651
47k
ACQ_INPUT
GND

#FLG610
PWR_FLAG
RAW_BPLUS

#FLG611
PWR_FLAG
RAW_BMINUS

#FLG612
PWR_FLAG
PACK_POS

#FLG613
PWR_FLAG
PROT_VDD

Primary protection per cell: OV 4.25V / UV 3.00V; inhibit OV-cell recharge. Charge/discharge overcurrent use VM and actual series resistance.
Q601/Q602 are common-DRAIN NMOS. Q603/Q604 are common-SOURCE PMOS: no body-diode bypass when OFF.
RUN typical falling cutoff ~6.63V, restart ~6.72V; OV cutoff ~8.6V. USB clamps RUN_BASE independently of MCU.
Raw B− returns only to protection/balance circuits. USB, charger, NTC and EEG use protected GND.
Hardware power-off is NOT galvanic or patient isolation. Remove electrodes before inserting USB.

ISOLATED USB PROTOTYPE — SYSTEM VALIDATION REQUIRED
YuEEG
Sheet: /07 / Raw 2S cell protection and fail-safe acquisition switch/
File: battery_protection.kicad_sch

Title: 07 / 2S protection and acquisition inhibit

Size: A3	Date: 2026-10-26	**Rev: D**
KiCad E.D.A. 9.0.5		Id: 9/13

CELL BALANCING: ONLY WITH VALID USB SOURCE AND RAW PACK ABOVE ~7.6V

RAW_BPLUS
R630
160R 1%
BAL_VC2

RAW_BPLUS
R633
100R 1%
BAL_VDD

BAL_VDD
C630
100n 16V
RAW_BMINUS

U603
BQ29209DRBR
BAL_VDD 7 VDD
BAL_VC2 1 VC2
BAL_VC1 2 VC1
BAL_VC1_CB 3 VC1_CB
BAL_ENABLE 6 CB_EN
OUT 8 BAL_OV_ALARM
CD 4 BAL_CD
GND 5
EP 9
RAW_BMINUS

RAW_MID
R631
260R 1%
BAL_VC1

BAL_VC2
C631
100n 16V
BAL_VC1

U605
TPS3700DDCR BAL UV
RAW_BPLUS 5 VDD
BAL_UV_SENSE 3 INA+
RAW_BMINUS 4 INB−
OUTA 1 RAW_OK
OUTB 6
GND 2
RAW_BMINUS

RAW_MID
R632
100R 1%
BAL_VC1_CB

BAL_VC1
C632
100n 16V
RAW_BMINUS

RAW_BPLUS
R660
1.81M 0.1%
BAL_UV_SENSE

BAL_UV_SENSE
R661
100k 0.1%
RAW_BMINUS

RAW_BPLUS
R662
1M
RAW_OK

RAW_BPLUS
C660
100n 16V
RAW_BMINUS

BAL_CD
C633
330n 16V
RAW_BMINUS

BAL_VDD
R634
330k
BAL_ENABLE

USB5
R663
2.2k
BAL_OPTO_A

U606
LTV−356T VALID USB
BAL_OPTO_A 1
CHG_PG 2
4 BAL_ENABLE
3 BAL_AND_SOURCE

Q607
2N7002 RAW UV GATE
BAL_AND_SOURCE
RAW_OK 1
3
2
RAW_BMINUS

TP601
RAW_BPLUS
RAW_BPLUS

TP602
RAW_MID
RAW_MID

TP603
RAW_BMINUS
RAW_BMINUS

TP604
BAL_OV_ALARM
BAL_OV_ALARM

#FLG620
PWR_FLAG
BAL_VDD

TP605
CHG_PG
CHG_PG

TP606
CHG_STAT
CHG_STAT

BQ29209 internal balancing: TI Fig9 resistor network, about 10mA; matched cells required. Not a cell equalizer for mismatched/old cells.
VC2 capacitor returns to VC1; VC1 capacitor returns to RAW B−. Do not return raw-domain nodes to system GND.
CB_EN active LOW: 330k pull-up disables by default. Optocoupler needs /PG LOW AND Q607 enabled by raw pack UV comparator.
Raw UV typical rising ~7.64V/falling ~7.53V; prevents balancing depleted cells. Optocoupler is domain translation, NOT patient isolation.
OUT is an alarm test point only; primary charge/discharge cutoff is independently provided by S−8252AAY.
J401 sequence and external NTC wiring must be verified before cells are plugged in. No standalone charger firmware is needed.

ISOLATED USB PROTOTYPE — SYSTEM VALIDATION REQUIRED
YuEEG

Sheet: /08 / Cell balancing with raw−domain UV qualification/		
File: battery_balance.kicad_sch		
Title: 08 / Cell balancing with raw−domain UV qualification		
Size: A3	Date: 2026−10−26	**Rev: D**
KiCad E.D.A. 9.0.5		Id: 10/13

HOST SIDE: NO ELECTRODE / SYSTEM GROUND CONNECTION

F701
1.5A hold PTC
HOST_RAW
HOST_5V

HOST_5V
C701
100n 16V
HOST_GND

USB_HOST_DP
R701
24R 1%
USB_UP_DP

USB_VDD1
C702
100n 16V
HOST_GND

#FLG703
PWR_FLAG
USB_VDD1

USB_HOST_DM
R702
24R 1%
USB_UP_DM

U701
ADuM4160BRIZ / enhanced creepage
HOST_5V
+3V3
USB_VDD1
USB_UP_DP
USB_UP_DM
USB_VDD1
USB_VDD1
VDD1
VBUS1
VBUS2
VDD2
PIN
UD+
UD−
PDEN
SPU
DD+
DD−
SPD
USB
GND1
GND1
GND2
GND2
Logic
+3V3
+3V3
USB_DOWN_DP
USB_DOWN_DM
+3V3
HOST_GND
GND

USB_DOWN_DP
R703
24R 1%
USB_MCU_DP

+3V3
C703
100n 16V
GND

USB_DOWN_DM
R704
24R 1%
USB_MCU_DM

D701
SMF5.0A
HOST_5V
HOST_GND

U702
TLV75533PDBVR HOST ONLY
HOST_5V
HOST_5V
IN
EN
OUT
GND
HOST_3V3
HOST_GND

HOST_5V
C704
1u 16V
HOST_GND

HOST_3V3
C705
1u 10V
HOST_GND

U703
TUSB320LAIRWBR / UFP GPIO
HOST_3V3
USB_CC1
USB_CC2
CC_VBUS_DET
HOST_GND
HOST_GND
VDD
CC1
CC2
VBUS_DET
PORT_UFP
EN_N
GND
OUT1
OUT2
OUT3
ID
ADDR_GPIO
CC_HIGH_N
CC_MODE2
HOST_GND

U704
SN74LVC1G04DBVR
HOST_3V3
CC_HIGH_N
VCC
NC
A
GND
Y
HOST_POWER_EN
HOST_GND

U705
TPS2553DBVR
HOST_5V
HOST_POWER_EN
HOST_ILIM
IN
EN
ILIM
GND
OUT
FAULT
HOST_DCDC_IN
HOST_GND

HOST_DCDC_IN
C708
4.7u 16V
HOST_GND

HOST_5V
C709
100n 16V
HOST_GND

#FLG701
PWR_FLAG
HOST_5V

#FLG702
PWR_FLAG
HOST_GND

HOST_5V
R705
887k 1%
CC_VBUS_DET

HOST_3V3
C706
100n 16V
HOST_GND

HOST_3V3
C707
100n 16V
HOST_GND

HOST_POWER_EN
R707
100k
HOST_GND

HOST_ILIM
R706
23.7k 1%
HOST_GND

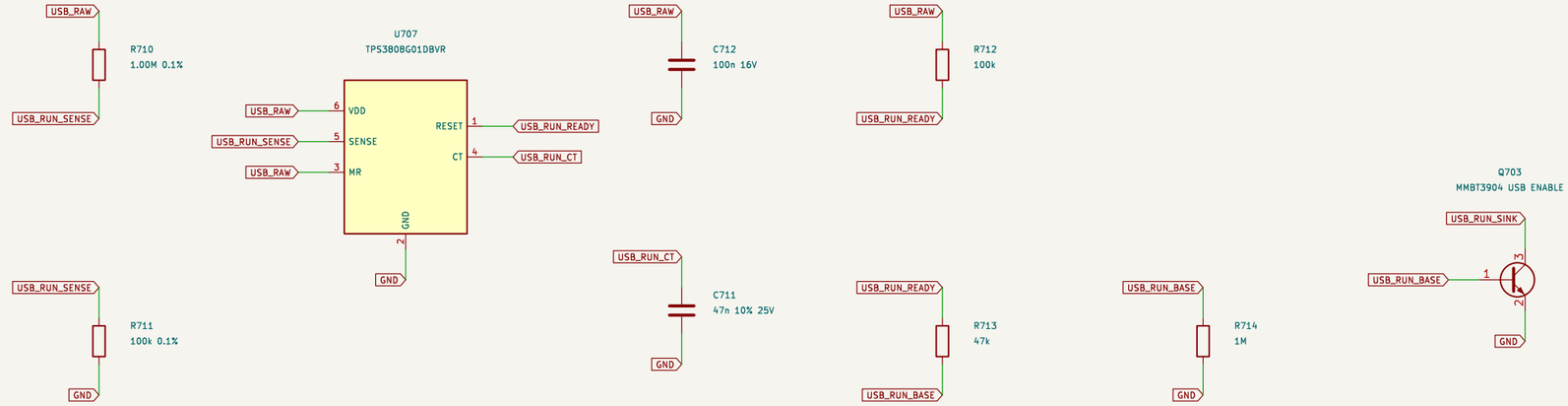
U706
THM 6-0511 / standard pinout
HOST_DCDC_IN
HOST_GND
HOST_GND
+VIN
−VIN
−VIN
+VOUT
−VOUT
USB_RAW
GND

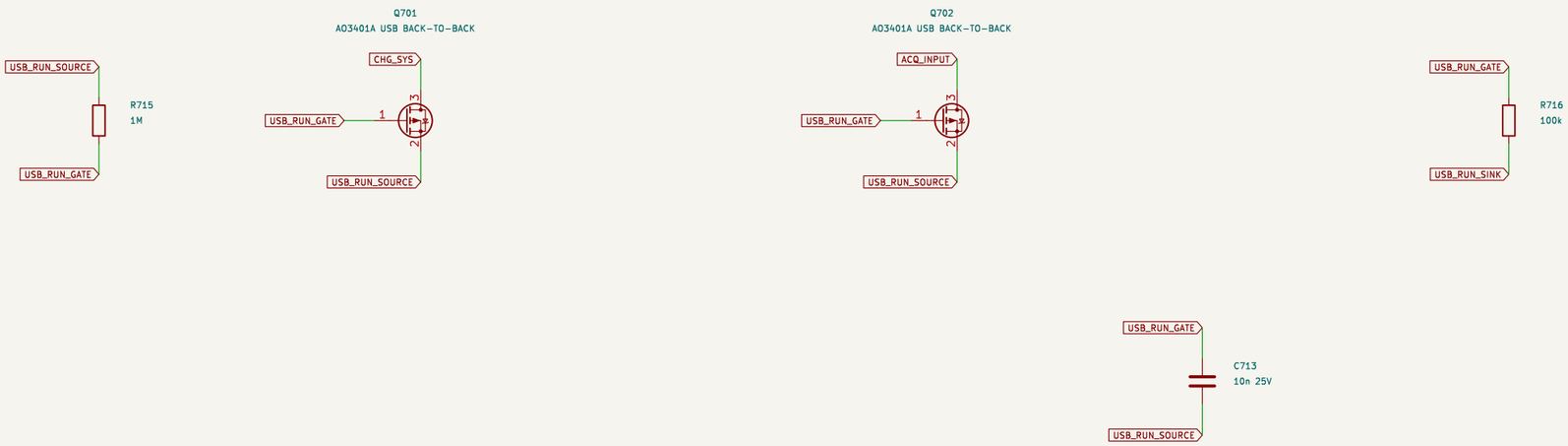
USB_RAW
C710
10u 25V X7R
GND

ADuM4160 BRIZ: full-speed 12Mbps; both speed pins HIGH. 24R series on all four USB lines.
Primary CC controller replaces external Rd resistors. OUT1 LOW only for advertised 1.5A/3A.
TPS2553 RILIM=23.7k: max about 1.17A before resistor tolerance; no PD negotiation.
THM6 standard pinout only, not -B1/W1 substitutions. Data isolator and converter ratings do not certify the assembled device.
No copper, vias, shield bond, external cable or screen supply may bridge HOST_GND and system GND.

ISOLATED USB PROTOTYPE — SYSTEM VALIDATION REQUIRED
YuEEG
Sheet: /09 / Isolated USB data and conditional isolated power/
File: isolated_usb.kicad_sch
Title: 09 / Isolated USB data and conditional isolated power
Size: A3 | Date: 2026-10-26 | **Rev: D**
KiCad E.D.A. 9.0.5 | Id: 11/13

USB DATA + WIFI MAY RUN TOGETHER; CHARGING IS DISABLED WHILE POWER IS ON





USB_RAW rising above about 4.455V must remain valid for about 0.2s before USB SYS path turns ON.
Existing battery path is inhibited immediately by Q606 sensing isolated USB_RAW. Delayed USB turn-on avoids a parallel charge path.
USB_RAW falling trips supervisor while Q606 still inhibits the battery; confirm break-before-make on bench.
POWER OFF charges battery; POWER ON supplies MCU/AFE from SYS and holds charger CE high.
Legacy/default-current USB: converter OFF, data works from battery with POWER ON. Type-C 1.5A/3A permits USB-powered flashing without battery.
USB-mode minimum pre-LDO headroom and switching noise need validation; external grounded instruments can defeat isolation.

ISOLATED USB PROTOTYPE – SYSTEM VALIDATION REQUIRED

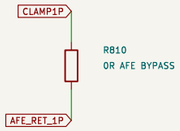
YuEEG

Sheet: /10 / USB SYS power path with delayed battery interlock/
File: usb_system_power.kicad_sch

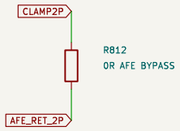
Title: 10 / USB SYS power path with delayed battery interlock

Size: A3	Date: 2026-10-26	**Rev: D**
KiCad E.D.A. 9.0.5		Id: 12/13

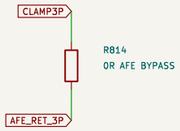
REMOVE BOTH BYPASS LINKS BEFORE DRIVING A CHANNEL FROM AN ACTIVE DAUGHTERBOARD



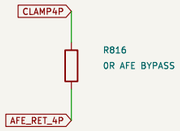
R810
OR AFE BYPASS



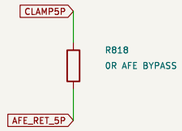
R812
OR AFE BYPASS



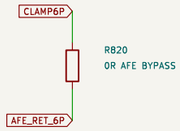
R814
OR AFE BYPASS



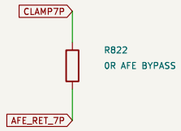
R816
OR AFE BYPASS



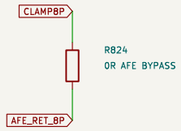
R818
OR AFE BYPASS



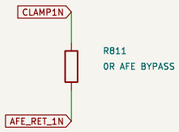
R820
OR AFE BYPASS



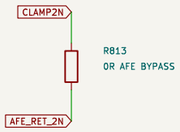
R822
OR AFE BYPASS



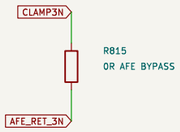
R824
OR AFE BYPASS



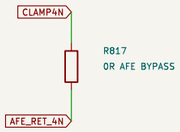
R811
OR AFE BYPASS



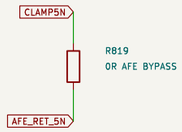
R813
OR AFE BYPASS



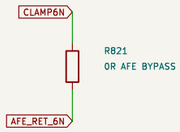
R815
OR AFE BYPASS



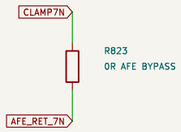
R817
OR AFE BYPASS



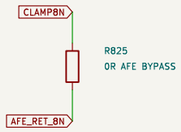
R819
OR AFE BYPASS



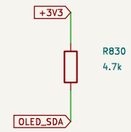
R821
OR AFE BYPASS



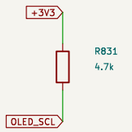
R823
OR AFE BYPASS



R825
OR AFE BYPASS



R830
4.7k

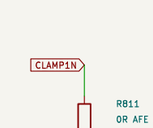


R831
4.7k

J801
AFE SEND / RETURN

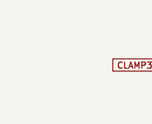
J802
AFE SEND / RETURN

J803
OLED 3V3 / GND / SDA / SCL

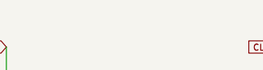


R840
2.2k





R841
2.2k

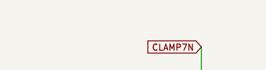




R842
2.2k



R843
2.2k



Default links preserve the existing passive input network. Optional daughterboard must buffer BOTH P/N with matched transfer functions.
Power budget for the two analog rails is limited; verify load, headroom, offset, input noise and stability before fitting an amplifier.
No automatic filter matching is claimed. Keep all external AFE and OLED supplies floating in the system domain.
OLED connector is 3.3V only; no display driver is enabled in current firmware. LEDs use static/slow GPIO, no PWM.

ISOLATED USB PROTOTYPE – SYSTEM VALIDATION REQUIRED
YuEEG

Sheet: /11 / Symmetric AFE insertion and status interfaces/
File: afe_expansion.kicad_sch

Title: 11 / Symmetric AFE insertion and status interfaces

Size: A3	Date: 2026-10-26	**Rev: D**
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